WEB DEVELOPMENT · 3RD SEMESTER

Week 03 · Block 3a

Styling with CSS

<web-development> · Hochschule Ansbach
Matthias Kellnhofer

After this block, you can…

- Explain what **CSS** is and how it relates to HTML
- Attach CSS **three ways** — and know why **external** wins
- Target elements with **type**, **class**, and **id** selectors
- Set your first properties: **color**, **fonts**, **text**
- Experiment with styles **live** in the dev tools

HTML gave your page **structure and meaning**.

CSS gives it a look.

Same markup — swap the stylesheet, and everything changes.

PART 1

What is CSS?

Cascading Style Sheets

Presentation, not structure

- **HTML** says *what things are*. **CSS** says *how they look*.
- Two separate layers — structure in one file, appearance in another
- Remember Block 2a: *"looks come from CSS, in Block 3"*? This is Block 3.

Keeping them apart is **separation of concerns**: you can restyle a whole site without touching the HTML — and the same markup can wear completely different looks.

Anatomy of a rule

```
p {
  color: navy;
  font-size: 18px;
}
```

- `p` — the **selector**: which elements this applies to
- `{ … }` — the **declaration block**
- `color: navy;` — one **declaration**: a **property**, a `:`, a **value**, a `;`

A **selector** plus its **block** is a **rule**. A stylesheet is just a list of rules.

PART 2

Attaching CSS

Three ways to attach CSS

```
<!-- 1. inline — right on the element -->
<p style="color: navy;">…</p>

<!-- 2. internal — a <style> block in the <head> -->
<style>
  p { color: navy; }
</style>

<!-- 3. external — a linked file (the standard) -->
<link rel="stylesheet" href="styles.css" />
```

All three set the same colour. Only one of them scales.

Use an external stylesheet

The `<link>` in the `<head>` is how real sites do it:

- **One file** styles every page — change it once, the whole site updates
- Keeps HTML about **structure**, CSS about **looks**
- The browser **caches** it — remember the `304` / "disk cache" from Block 1? That's your stylesheet loading instantly on the next page

> Inline and internal are fine for a five-second test. For anything real, reach for an external file.

PART 3

Selectors

The three you'll use most

```css
h1      { }   /* type  — every <h1> on the page   */
.note   { }   /* class — any element class="note" */
#logo   { }   /* id    — the one element id="logo" */
```

```html
<h1>Title</h1>
<p class="note">A tip.</p>
<img id="logo" src="logo.svg" alt="Logo" />
```

- **type** — all elements of that kind
- **class** (`.`) — **reusable**: put `class="note"` on as many elements as you like
- **id** (`#`) — **unique**: one per page, used sparingly

Combine them

```
h1, h2 {  }   /* grouping   — applies to both h1 and h2 */

nav a  {  }   /* descendant — an <a> anywhere inside <nav> */
```

- **Grouping** with a comma — give several selectors the same rules
- **Descendant** with a space — target elements *inside* another

When two rules touch the same element, which one wins? That's the **cascade** and **specificity** — the whole of Block 3b.

PART 4

First Properties

Enough to make things happen

```
body {
  font-family: system-ui, sans-serif;
  color: #14161b;
  background-color: #f7f7fb;
}
h1 {
  font-size: 40px;
  text-align: center;
}
```

Your starter set: `color` · `background-color` · `font-family` · `font-size` · `font-weight` · `text-align`.

Colours, units, typography and spacing get the full treatment in Block **3c** — this is just enough to start.

PART 5

Style It Live

The Styles pane is your sandbox

Everything CSS is editable right in the browser:

- **Elements** panel → pick an element → the **Styles** pane on the right
- Edit values, toggle declarations on/off, **add a new rule** — applied instantly, no reload
- See exactly which rules hit the selected element

> Nothing here is saved — it's a scratchpad. Experiment until it looks right, then copy the winning declarations into your `styles.css`.

Recap

- CSS is the **presentation** layer: HTML = structure, CSS = looks
- A **rule** = `selector { property: value; }`
- Attach it three ways — prefer an **external stylesheet** (reusable, cached)
- Select by **type**, **class** (`.`), and **id** (`#`)
- Prototype **live** in the dev tools Styles pane, then save it

<EXERCISE> · 45 MIN

Your turn 👉

Give a plain HTML page a look — with your own external stylesheet.

1. **Link** an empty `styles.css` from the page's `<head>`
2. Style with all three selectors: **type**, **class** (`.`), and **id** (`#`)
3. Set **`color`**, **`background-color`**, **`font-family`**, **`font-size`**, **`text-align`**

Starter: `exercises/week03/3a/` — open `README.md`. The HTML (with classes & ids ready to target) and a `styles.css` are waiting.